AALBORG UNIVERSITY
DENMARK

4 LEVERAGE

ESSENCE – CHAPTER 7

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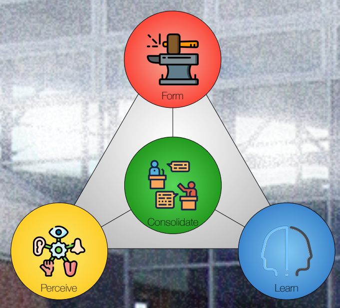
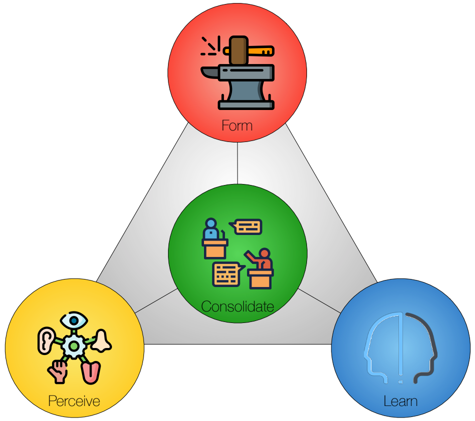


Agenda

- Exercise 3 Follow-up.
- Chapter 7: Leverage.
  - Contribution.
    - Means.
    - Transactions.
  - Leverage.
    - Leverage categories.
    - PCRT idea evaluation.
  - Leverage scenarios.
- Next time:
  - Exercise 4: Leverage Scenarios.
  - Lecture 5: Solution.

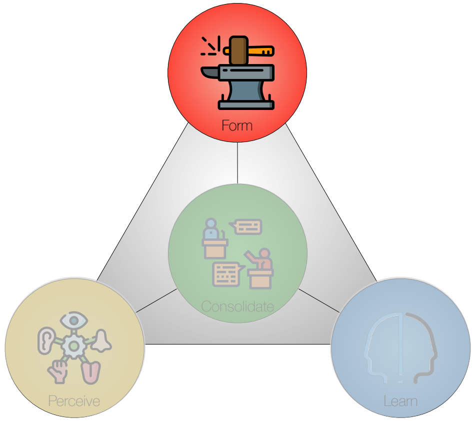


Follow-up: Exercise 3

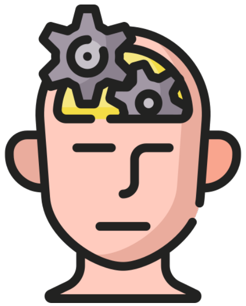
Exercise 3 (Problem Scenarios)

Use the results from Exercise 2 and the PSC Template in this exercise.

- Describe the context type of your project as defined in Table 6.1.
- Develop axes for Problem Scenarios as described in Section 6.3 (Tables 6.2, 6.3, 6.4, and 6.5). Choose two axes for the Problem Scenarios.
- Develop Problem Scenarios as illustrated in Figure 6.1. Explain the problem found in each quadrant.
- Choose one problem to work with from now on. You can choose one quadrant or combine several if relevant. Express this problem in one sentence in the PSC problem cell (Section 3.1).
- Outline how this problem manifests itself in the PSC manifestations cell (Section 3.3).
- Outline the Outer Environment of your project and enter these elements in the Outer Environment cell (Section 3.2).



Form

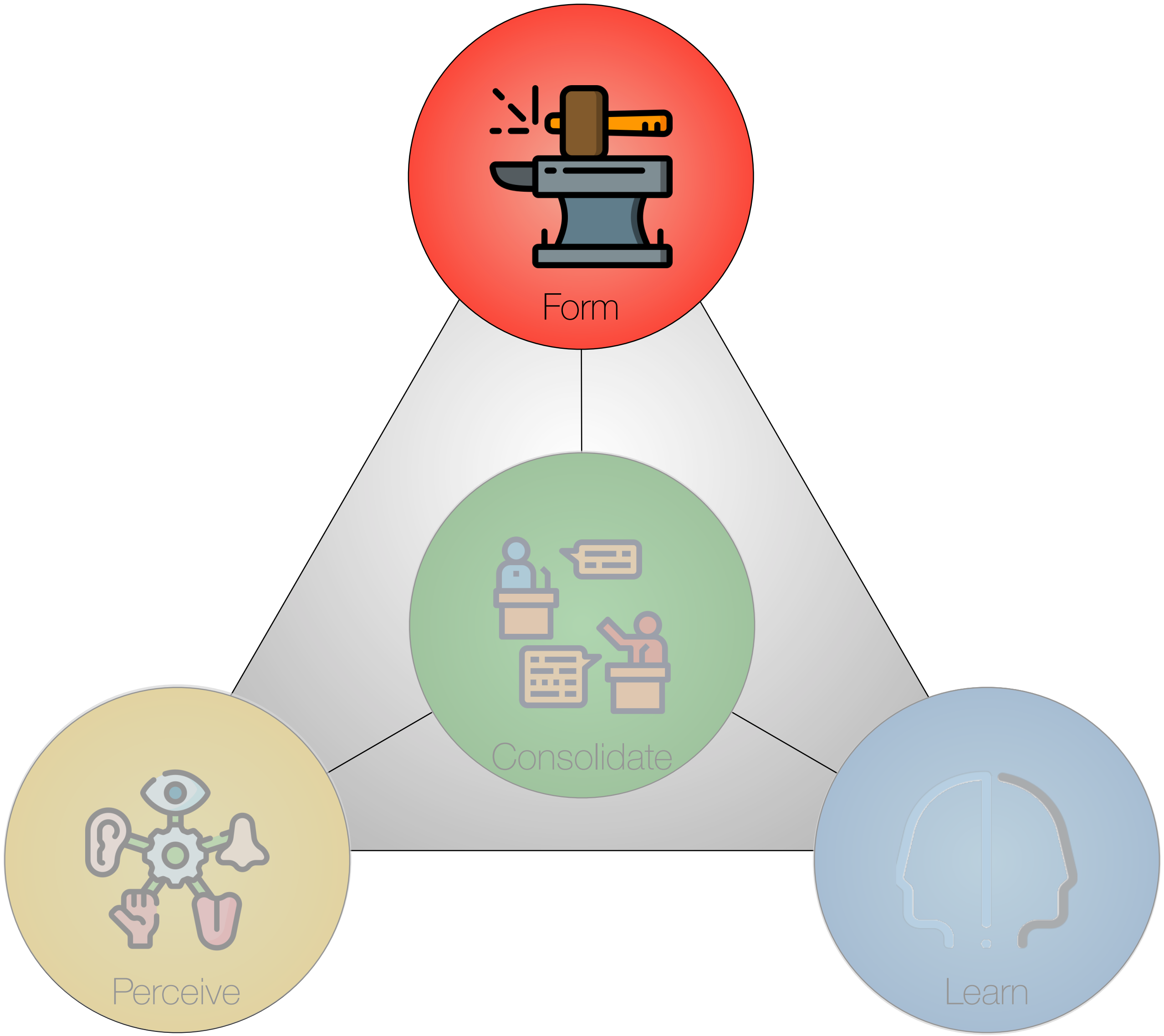


Things are not always as they seem; the first appearance deceives many.

Plato, *Phaedrus* (15 BC - 50 AD)

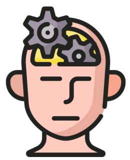
Four Core Activities

- *Perceive* – understanding the problem and its context.
- *Form* – designing the constructive parts of a solution.
- *Consolidate* – integrating the design contributions into a clear whole that addresses the situation.
- *Learn* – appraising the design.



Icons from Flaticon.com

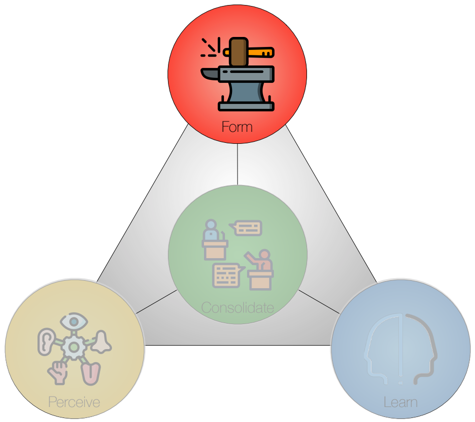
Three Levels of Abstraction



Rationale – the logical basis that makes actions meaningful and helps reason about strategy and tactics for solving the overall problem.

Strategy – the master plan for solving the overall problem, including the scope of the problem, the key components for building the solution, and qualifications regarding limitations and constraints that may affect the utility or acceptability of a solution.

Tactics – actions planned in accordance with the strategy to achieve specific ends.



Deweyan Pragmatism

Every idea originates as a suggestion, but not every suggestion is an idea. The suggestion becomes an idea when it is examined with reference to its functional fitness; its capacity as a means of resolving the given situation.

John Dewey (1859-1952)

Means – Material and Procedural

- *Means* are instruments used to attain an end – something useful in achieving a result.
  - *Material means* – including observed data, facts and artifacts – combined serve to create a resolved situation.
  - *Procedural means* – including operations, plans, requirements, and test cases – serve to determine how and if such a resolved situation is attained.

ClusterDetect material means: Number of people, legal limitations, sensors …
ClusterDetect procedural means: PSC, experiments, spikes, prototyping, …

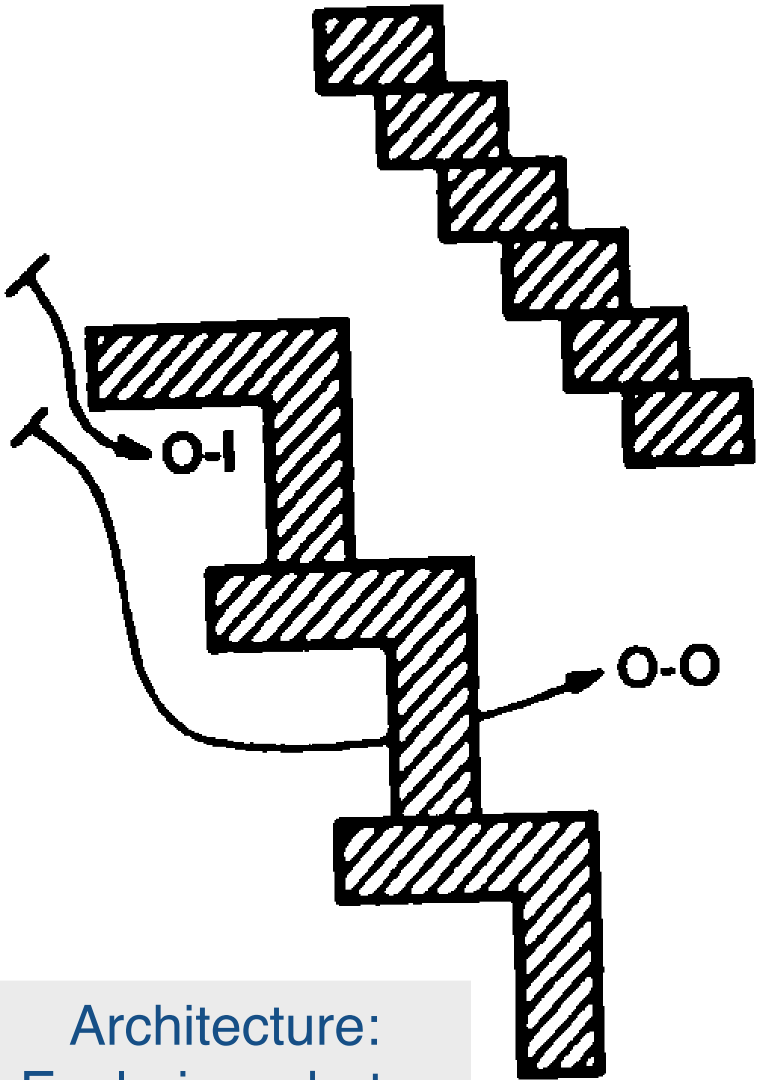
Ideas

Ideas are anticipated consequences (forecasts) of what will happen when certain operations are executed under and with respect to observed conditions.

To Dewey, ideas are similar to hypotheses

John Dewey (1859-1952): *Logic: The Theory of Inquiry* (1938) p109.

Transaction

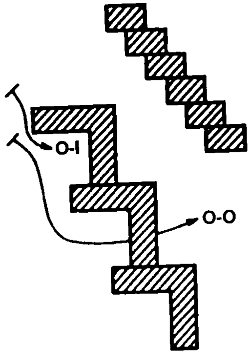


The inquirer does not stand outside the problematic situation like a spectator; he is *in* it and in *transaction with* it

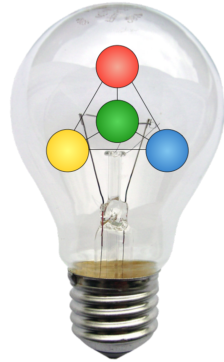
[…] an inquirer, in transaction with the materials of a situation, encounters a surprise in the form of "back-talk" that momentarily interrupts action, evoking uncertainty. (Schön, 1992b, page 125).

Architecture: Exploring what a school could be

Schön, D. A., & Wiggins, G. (1992). *Kinds of seeing and their functions in designing*. Design Studies, 13(2), 135-156.



Using Essence on the School Case



Core Concern	Situation	Contribution	Solution	Valuation
RATIONALE: Why?	(P) PROBLEM A space for good teaching A social space A public space	(L) LEVERAGE Modern teaching facilities Grade-adapted O-I exterior Inviting O-O exterior	(R) RESOLUTION • PROSPECT • WARRANT • BACKING	(CR) CRITERIA FOR RATIONALE
STRATEGY: What?	(O) OUTER ENVI-RONMENT Users of classrooms Users of social space Users of public space	(I) INNER ENVI-RONMENT Internal facilities Playground for grades Adaptation to the exterior	(Q) QUALIFICATION • RESERVATION • REBUTTAL	(CI) CRITERIA FOR STRATEGY
TACTICS: How?	(M) MANIFESTA-TIONS Lessons Breaks Public uses	(C) CAPABILITIES Teaching and cloakroom Leisure for grades Public access	(V) VALUE	(CV) CRITERIA FOR TACTICS

Red: First perceptions Blue+Green: After transacting with the problem

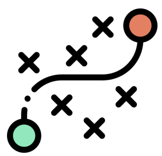
Continuity

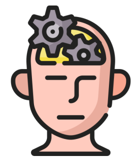
- The means employed to resolve a situation are subjected to critical reflection in the duration of the project.
- They are constantly contrasted with realities through transaction, and they are changed in accordance with experience.
- Means therefore are both a basis for and a result of transactions.

There is continuity in inquiry. The conclusions reached in one inquiry become means, material and procedural, of carrying on further inquiries

John Dewey (1859-1952): *Logic: The Theory of Inquiry* (1938) p140.

Form Terminology

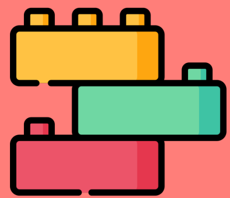




Capabilities

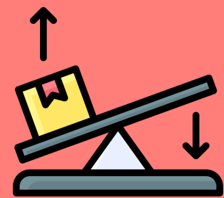
Capabilities are the features that are designed to address manifestations.

Inner Environment



The inner environment is what provides the capabilities.

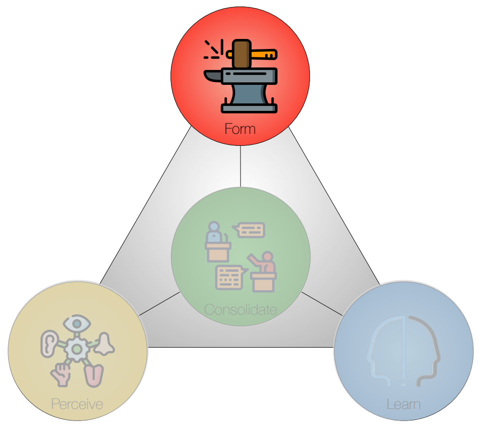
Leverage



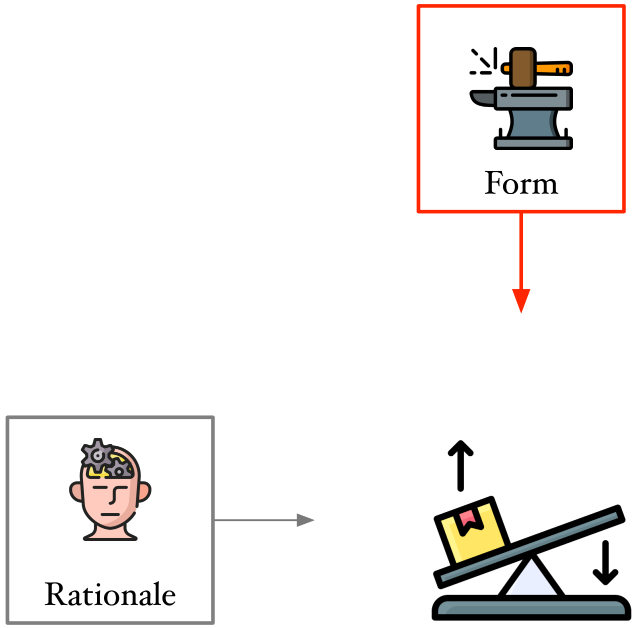
Leverage points allow for the most valuable capabilities.

Illustration: ClusterDetect

	Perceive	Form	Consolidate	Learn
Rationale	**Problem** In crowded areas, safety could be compromised	**Leverage** Technology: AI [DFNet] Components: [Thermal camera] Information: Problem spots Human resources: [On-the-ground staff]	**Solution** Prospect: ClusterDetect reduces congestion and disruption Warrant: Densely populated areas are common and dangerous Backing: The Contribution is effective and inexpensive	**Horizon** The project creates strengths and opportunities - The Problem is understood and generic - The Leverage is future-proof - The Prospect solves the Problem, - The Warrant suggests growth - The Backing suggests motivation to adopt
Strategy	**Outer Environment** External service: [Access control] Implements: Venue information system Repository: Traffic information People: [Informants]	**Inner Environment** Density monitor [Thermal?] Panic and disruption locator General modules for on-the-ground staff and visitors [Customized module for on-the-ground staff]	**Evolvability** Diffusibility: ClusterDetect is highly specialized and has low diffusibility towards new markets Adoptability: ClusterDetect is limited to the current market, but it can address new needs and thereby strengthen existing market positions	**Potential** The design has strategic potential: - Outer Environment is generalized - Inner Environment is near-decomposable - The Evolvability is limited but non-critical
Tactics	**Manifestations** Potential crowding Places of congestion Potential starts of panic Potentially disruptive actions	**Capabilities** Identify and handle crowds Identify and handle congestion Identify and handle panic Identify and handle disruption	**Merit** Value: Locates impediments. Finds and handles crowds, congestion, panic, and disruption Reservation: Requires calibration and clear sight Rebuttal: Many users will accept these limitations in a low-cost system	**Mission** ClusterDetect solves the Problem: - The Manifestations are essential - The Capabilities are efficient and effective - The Merit offered ensures sustained use



Leverage

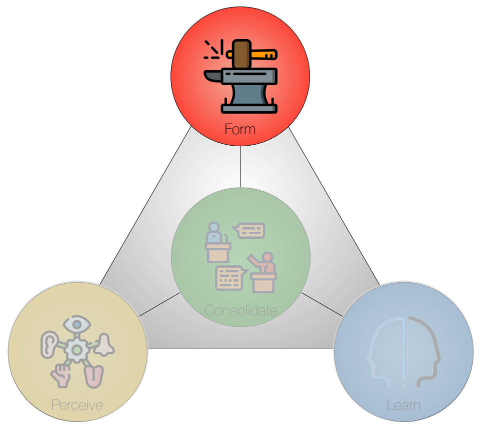


What is Leverage?

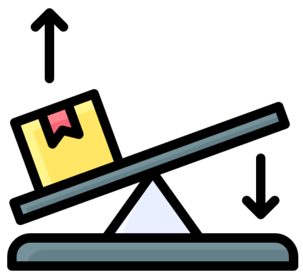
Leverage is short for leverage points – technologies, components, information, or human resources that could be keystones in an excellent solution.

A leverage point might lead to a solution with more substantial features, lower cost, greater flexibility, robustness, or any other quality to make a design stand out.

Proposed leverage points are ideas; they are forecasts pointing to possible effects on the project or the product. These forecasts are based on examinations of their capacity to help solve the problem.



Leverage Categories & Scenarios



Four Leverage Categories

Categories	Characteristics
Technology	Scientific knowledge, machinery, and equipment on which a design might be based.
Components	Rare equipment, sensors, or actuators used actively in building a solution and accessed via crafted interfaces.
Information	Used for collecting, storing, disseminating, and/or aggregating information to be accessed via crafted interfaces.
Human resources	Specific users, categories of users, experts, authorities, social or professional networks, organizations, or others with particular roles or functions designed to build a solution. Cyberhuman systems are prominent examples of systems where internal people are leverage points.

Crafted interfaces indicate that items of these categories must be used as an integrated part of the design

Leverage Ideas for ClusterDetect

Categories	Characteristics
Technology	*Trilateration* using Bluetooth signal strength from beacons and smartphones to locate individuals and calculate group density.
Components	*Smart cameras* equipped with body recognition software could detect individuals, and by combining images from multiple angles, it would be possible to estimate group density in a particular location.
Information	Humans radiate body heat; *thermal cameras* can use radiation intensity to create heat maps and estimate the density of groups.
Human resources	*Trained staff* could monitor areas via visual inspection. Such staff could coordinate across enclosures and reduce the population density areas in time before crowding becomes a concern.

Qualitative PCRT
(Single Idea Analysis)

<table>
<tr><td rowspan="4">Trilatera-
tion
(Technology)</td><td colspan="3">POWER</td></tr>
<tr><td colspan="3">Medium. The technology is non-intrusive, and almost everybody wears equipment emitting Bluetooth signals. The accuracy in location equipment does, however, seem relatively low. Bluetooth range is relatively short, and this might limit where this technology is helpful. The potential for this approach might have reached its peak, and further developments in accuracy seem unlikely.</td></tr>
<tr><td>COST</td><td>RISK</td><td>TIME</td></tr>
<tr><td>Low since this technology can use cheap and standard equipment. Development costs should also be low in teams with the relevant competencies.</td><td>Moderate to high. Bluetooth signal strength varies based on the brand, models, and other factors related to the equipment present in the area. Such variations significantly impact accuracy.</td><td>Low since the complexity of design and implementation will likely be low.</td></tr>
</table>

- *Power* (how powerful might this option be?).
- *Cost* (how do the costs match the gains? Will the gains be worth it?).
- *Risk* (how much risk might be involved?).
- *Time* (how might this affect the time to complete?).

Quantitative PCRT
(Comparative Idea Analysis)

Leverage point Ideas	Assessment						Sum	Total
Trilateration *(Technology)*	Power						5	1
	Cost	1	Risk	4	Time	1	6	
Smart cameras *(Components)*	Power						7	4
	Cost	1	Risk	1	Time	1	3	
Thermal cameras *(Components)*	Power						8	2
	Cost	3	Risk	1	Time	2	6	
Trained staff *(Human resources)*	Power						3	6
	Cost	6	Risk	1	Time	2	9	

- *Power* (how powerful might this option be?).
- *Cost* (how do the costs match the gains? Will the gains be worth it?).
- *Risk* (how much risk might be involved?).
- *Time* (how might this affect the time to complete?).

Leverage Scenarios

Alternative Interpretations of the Leverage

Leverage	Vertical Axis
Who	Who will be part of the contribution? Who are the operators?
Why	Why is this a contribution? Is it bespoke or generic? Is it contextual, universal, preventive, ameliorative, mitigating, compensating, active, passive, supportive, ...?

Vertical Axis

Leverage Scenarios
ClusterDetect case

- *Who* will operate the system? One option could be safety assurance specialists working at the venue. Another option could be on-the-ground staff handling crowd building during an event.
- *Why* is this a contribution? Will the system help prevent dangerous situations, or will it help control disasters? Will it address a specific situation or a broader range of problems? Will it only be helpful concerning disasters, or will it have wider applications?

Leverage Scenarios 2

Leverage	Horizontal Axis
What	What are the inner characteristics of the contribution? Is it flexible, decomposed, integrated, platform-based, open, closed, standardized, …?
Where	Where will the contribution be used? Where will it have an effect? What part of the problem is targeted?
When	When will the contribution be used? When will the effects be seen (immediately, delayed, or gradually)? Will the contribution be used according to a plan, on demand, constantly, repeatedly, only once, …?
How	How will it work? Will the contribution work directly, indirectly, or via network effects? Will it be active or passive, intrusive or non-intrusive, proactive or reactive, switched or always-on?
How much	How much will the contribution cost? Can the expenses be quantified in terms of investment and operations? Can the savings be quantified in terms of time, money, manpower, risk, …?

Horizontal Axis

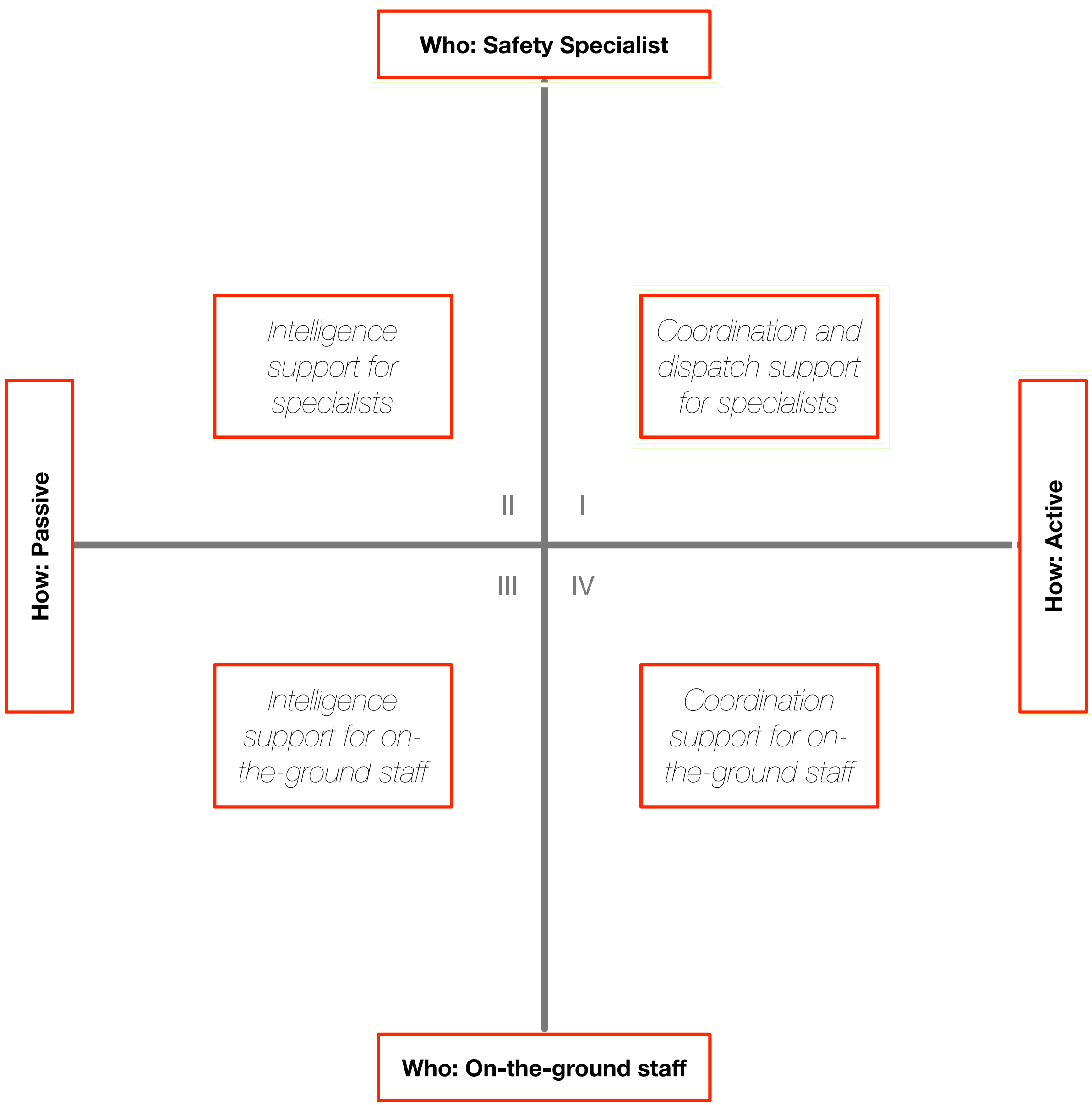
Leverage Scenarios
ClusterDetect case

- *What* characterizes the contribution? Is the system highly integrated for efficiency, or is it decomposed for flexibility and a longer lifespan?
- *Where* will the system have an effect? Will it work directly with people, or will it work with the location's infrastructure?
- *When* will the system be used? During planning and prevention work or in critical situations?
- *How* will the system work? Will it passively collect and present information, or will it actively coordinate or control interventions?
- *How much* will the system cost? Will cost considerations mainly focus on investments or operations? Will savings come mainly from reducing manpower costs or from minimizing risks?

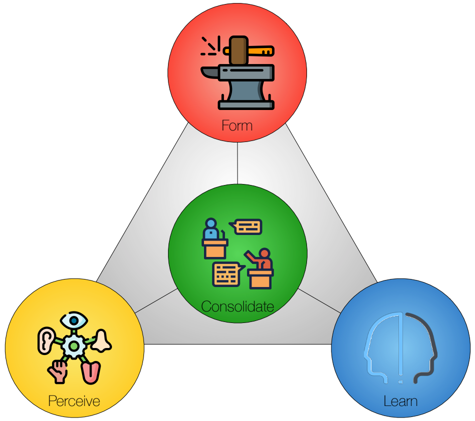
Horizontal Axis

Leverage Scenarios

ClusterDetect



Example based on one among many alternative axes



Next Time

Next Exercise and Lecture

- Exercise 4:
  - Leverage Scenarios.
- Lecture 5:
  - Chapter 8 in Essence: *Solution*.

Exercise 4 (Leverage Scenarios)

Use the results from Exercise 3 in this exercise.

- Suggest leverage points for your project. Use the categories in Table 7.1 for inspiration, or develop your own.
- Evaluate and select leverage points. Use PCRT-analysis as shown in Table 7.3 and/or Table 7.4.
- Develop axes for the Leverage Scenarios as described in Section 7.3 (Tables 7.5, 7.6, 7.7, and 7.8).
- Develop Leverage Scenarios as illustrated in Figure 7.1.
- Explain the Leverage Scenarios you find in each quadrant.
- Choose one scenario to work with from now on. You can choose one quadrant or combine several if relevant.
- Discuss the most important leverage points you plan to use in your project. Enter these leverage points in the Leverage cell (Section 3.6).
- Discuss the capabilities obtainable through these leverage points. Enter these capabilities in the Capabilities cell (Section 3.4).
- Discuss the Inner Environment of your design. Enter the key modules in the Inner Environment cell (Section 3.5).